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Exact Precision Thresholds for Frobenius Polynomial Reconstruction from Quantum Trace Records

A reply to the RT Sector Connection comment on "Quantum Trace Estimation and Frobenius Polynomial Reconstruction", with independent verification

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Abstract

A comment on our paper (Cooperman & Petrunina, rev. 26 September 2026), supported by the document *RT Sector Connection: Spectral Reconstruction from Quantum Trace Records* (H. Hajizadeh, October 2026) and a reproducibility package, makes five statements about recovering the integral Frobenius numerator $L(T)$ from noisy power-trace records. We set each statement against what our paper actually states, and then assess it. The package was replayed in full and reproduces byte-identically; all assessments rest on independent code. We find that the commentator's central theorem is correct: on the class of integer Weil polynomials of genus g, the exact worst-case identification threshold is $\delta_{cong} = \min_{1\le k\le g} k/(2Q^{k/2})$, attained by a congruence-aware decoder. This answers the precision part of our Research Question 8.1 and does not contradict any statement of our paper. The commentator's claim that Newton-then-round "loses about $g \cdot 2^{2g} \cdot H_{2g}$" is not correct as stated: that factor measures the looseness of our *sufficient* bound. Computing the true worst-case threshold of the operator gives losses of 1, ≈9 and ≈23–47 for $g = 1,2,3$, against stated values of 6, ≈67 and ≈470.

Validation code. Every result in this white paper can be checked with Python validation scripts, which are published on GitHub at:

https://github.com/teknoquants/Exact-Precision-Thresholds-for-Frobenius-Polynomial-Reconstruction-from-Quantum-Trace-Records.

The scripts are written independently and do not import code from the commentator's package. They:

- enumerate the admissible Weil-polynomial classes exactly and confirm the threshold δ_cong by brute force (§3);
- verify the witness pairs for genus 1 to 12 in exact arithmetic;
- recount the anchor curve $y^2 = x^5 - x$ over F_3 (§9);
- compute the true worst-case thresholds of the Newton-then-round operators in Table 1 (§5);
- run each failure witness through the operator at 1.001× and 0.999× the threshold (§5.1).

The repository includes the JSON outputs and a README with run instructions (Python 3.13, NumPy, SciPy, SymPy; about 4 minutes in total), so any reader can reproduce the findings independently.

1 Purpose and sources

This paper is written for readers who wish to compare the original research with the comment and to see the evidence behind each conclusion. For every statement in the comment we give, in order:

- **Our paper**: what the original research states, with section references.
- **The commentator**: what the comment states and where its argument comes from in the RT document.
- **Assessment**: who is correct, and the evidence.

The sources are:

- [CP] R. Cooperman and Petrunina, *Quantum Trace Estimation and Frobenius Polynomial Reconstruction*, revised 26 September 2026.
- [RT] H. Hajizadeh, *Return Theory (RT) Sector Connection: Spectral Reconstruction from Quantum Trace Records*, October 2026, with reproducibility package dated 8 October 2026.

The comment as received reads:

> *"I analysed your trace-estimation problem. Under integrality and the functional equation, the exact worst-case precision for recovering L(T) is δ = min over k of k / (2 Q^(k/2)), for every genus g and every Q. A congruence-aware decoder attains it, so Newton-then-round loses about g · 2^(2g) · H_(2g), in the reconstruction rather than in the record. This bears on your RQ 8.1."*

2 Setting and notation

Let $L(T)$ be an integer polynomial of degree $2g$ with constant term 1,

$$L(T) = \sum_{k=0}^{2g} b_k\, T^k\,, \qquad b_0 = 1, \qquad b_{2g-k} = Q^{g-k} b_k \quad (0 \le k \le g), \tag{1}$$

all of whose reciprocal roots $\alpha_1, \dots, \alpha_{2g}$ have modulus $\sqrt{Q}$. The set of all such polynomials is denoted $W_g(Q)$ [RT, Def. 3.1]. By the Weil bounds it contains the L-polynomials of all genus-g curves over F_Q, but it is a strictly larger class. The power sums $s_j = \sum_i \alpha_i^j$ are integers, and the record consists of the normalised traces

$$p_j = \frac{s_j}{Q^{j/2}}, \qquad j = 1, \dots, m, \qquad g \le m \le 2g, \tag{2}$$

each observed as $\hat{p}_j$ with adversarial error $|\,\hat{p}_j - p_j\,| \le \delta$. Coefficients and power sums are linked by Newton's identities,

$$k\, b_k = -s_k - \sum_{j=1}^{k-1} s_j\, b_{k-j}\,, \qquad k = 1, \dots, 2g. \tag{3}$$

The **worst-case identification threshold** of a class is $\delta_{sep} = \frac{1}{2}\min_{L \neq L'} \max_{j \leq m} | p_j(L) - p_j(L') |$: below it some method recovers every member; at or above it no method can, because the midpoint record of the closest pair is admissible for both members [RT, Lemma 6.1]. The **threshold of an operator** is the largest δ for which that particular operator recovers every member.

The operators discussed are:

- A_{all}: Newton reconstruction on $\hat{p}_1, \ldots, \hat{p}_{2g}$, then rounding of every $Q^{k/2}\hat{c}_k$ (the route analysed in [CP]).
- A_{FE}: as A_{all}, but rounding only $b_0, \ldots, b_g$ and completing by the functional equation (suggested in [CP, §6.1]).
- D: the congruence-aware decoder of [RT, Def. 3.2].

3 Statement 1: the exact threshold

> *"Under integrality and the functional equation, the exact worst-case precision for recovering L(T) is δ = min_k k/(2Q^{k/2}), for every genus g and every Q."*

Our paper. We did not derive this threshold, and we made no claim that contradicts it. Three statements of [CP] are relevant.

- **Theorem 3.2** gives a sufficient perturbation bound for Newton reconstruction of an arbitrary unitary spectrum of dimension d:

$$\max_{0 \leq k \leq d} | \hat{c}_k - c_k | \leq 2^d (e^{\delta H_d} - 1), \qquad H_d = \sum_{j=1}^{d} \frac{1}{j}. \tag{4}$$

We describe it as "a sufficient worst-case bound; the exact conditioning can be better on particular inputs."

- **Proposition 6.1** gives a sufficient precision for integer rounding. We noted that it "is not necessary in every arithmetic family", and that by the functional equation only $b_0, \ldots, b_g$ need independent recovery.
- **Theorem 3.4** proves that no algorithm recovers all coefficients uniformly from inverse-polynomially accurate traces of arbitrary unitaries. We stated that it "does not apply automatically to the integral, conjugate-paired spectra of a specified arithmetic family", which "requires separate analysis".

The commentator. The formula is the RT document's own result. It is assembled in three steps.

1. **Congruence [RT, Thm 7.1].** In the Newton identity (3), once $s_1, \ldots, s_{k-1}$ are fixed and every b is an integer, s_k is determined modulo k. Two distinct members of $W_g(Q)$ first differ at some $k \leq g$, and there $| \Delta s_k | \geq k$, so their records differ by at least $kQ^{-k/2}$. Hence $\delta_{sep} \geq \delta_{cong}$, where

$$\delta_{cong}(g, Q) = \min_{1 \leq k \leq g} \frac{k}{2Q^{k/2}}. \tag{5}$$

   A decoder that rounds each s_k within the residue class fixed by integrality succeeds whenever $\delta < \delta_{cong}$.

2. **Matching witnesses [RT, Lemma 7.2].** Two explicit families of admissible pairs lie at record distance exactly $2\delta_{cong}$:

$$L_0 = \sum_{i=0}^{g} Q^i T^{2i}, \qquad L_1 = L_0 - T\sum_{i=0}^{g-1} Q^i T^{2i}, \tag{6}$$

$$M_0 = 1 + Q^g T^{2g}, \qquad M_1 = M_0 + T^g. \tag{7}$$

   The power sums of L_1 and L_0 differ by 1 at every $j \le 2g$ (binding at $k = 1$); those of M_1 and M_0 differ by $-g$ at $j = g$, by g at $j = 2g$, and by 0 elsewhere (binding at $k = g$).

3. **Every k [RT, Thm 7.3].** The function $k \mapsto log\ k - \frac{k}{2} log\ Q$ is concave, so $kQ^{-k/2}$ attains its minimum over $1 \le k \le g$ at $k = 1$ or $k = g$, where the two families apply. Therefore $\delta_{sep}(W_g(Q)) = \delta_{cong}(g, Q)$ for every $g \ge 1$ and every integer $Q \ge 2$.

The ingredients — Newton's identities, integrality of the L-polynomial, and the functional equation — are classical. To our knowledge, their combination into an exact two-sided threshold for the trace-record model first appears in [RT].

> **Assessment. The commentator is correct, and there is no conflict with our paper. Theorem 7.3 of [RT] is the "separate analysis" called for in our caveat to Theorem 3.4, and it confirms our remark after Proposition 6.1 that the sufficient condition is not necessary in arithmetic families.**

Independent verification. Using code that imports nothing from the commentator's package:

- $W_g(Q)$ was enumerated with an exact algebraic root test (reduction to the real trace polynomial, exact factorisation and real-root isolation). The class sizes agree with [RT], e.g. $| W_2(4) | = 101$, $| W_3(3) | = 677$.
- Brute-force computation of δ_{sep} over all pairs gives $\delta_{sep} = \delta_{cong}$ in all 16 cases tested ($g = 1$: $Q \le 49$; $g = 2$: $Q \le 9$; $g = 3$: $Q \le 3$).
- Both witness families pass exact integer checks (integrality, functional equation, stated power-sum differences, distance $2\delta_{cong}$) for $g = 1, \dots, 12$ and ten values of Q up to 128; the largest deviation of a root modulus from $\sqrt{Q}$ in the normalised chart is 4×10^{-15}.

Two qualifications apply to the wording of the comment:

- *The class is $W_g(Q)$, not the set of curves.* For L-polynomials of genus-g curves the threshold is proved only for $g = 1,2$ [RT, Thm 7.5]; for $g \ge 3$ it is open, as [RT] itself states.
- *"Precision" is adversarial sup-norm precision on the normalised traces,* with strict inequality $\delta < \delta_{cong}$.

4 Statement 2: the congruence-aware decoder

> *"A congruence-aware decoder attains it."*

Our paper. [CP] analyses Newton reconstruction followed by rounding (Proposition 6.1, Theorem 4.1). It does not propose a decoder that enforces integrality before the Newton step.

The commentator. The decoder D [RT, Def. 3.2; Thm 7.1(ii)] rounds s_1 to an integer; for each $k = 2, \dots, g$ it rounds s_k to the nearest integer in the residue class $s_k \equiv -\sum_{j<k} s_j b_{k-j} \pmod{k}$; it then completes L by exact Newton and the functional equation.

Assessment. The commentator is correct. D is correct by construction whenever $\delta < \delta_{cong}$. This is a genuine improvement over the operator analysed in our paper.

Verification. In the replay, D is exact on every member of all 43 computed classes (20,919 polynomials) and recovers L in all 189,000 random records with $\delta < \delta_{cong}$.

5 Statement 3: the loss of Newton-then-round

> *"...so Newton-then-round loses about g · 2^(2g) · H_(2g)."*

Our paper. The bounds of Theorem 3.2 and Proposition 6.1 are stated as sufficient. [CP, §1.3] states that "no optimality … claim follows." Our paper does not state the exact worst-case threshold of Newton-then-round and does not claim that its bound equals it.

The commentator. The factor is the ratio of δ_{cong} to our sufficient bound,

$$\delta_{A_{FE}}^{suff} = \frac{ln(1+2^{-(2g+1)}Q^{-g/2})}{H_{2g}}, \qquad \frac{\delta_{cong}}{\delta_{A_{FE}}^{suff}} \approx g \cdot 2^{2g} \cdot H_{2g}. \tag{8}$$

The header of [RT, Table 1] says so explicitly ("their sufficient bounds from [Thm 3.2, Prop. 6.1]"). The abstract of [RT], its Corollary 7.6 and the comment itself, however, present this ratio as the loss of the operator.

Assessment. The commentator's statement is incorrect as worded. A sufficient bound underestimates an operator's tolerance; dividing δ_{cong} by it measures the looseness of the bound, not the loss of the operator. The word "so" does not follow, because an exact threshold for decoder D carries no information about the exact threshold of a different operator.

5.1 The true worst-case threshold

To settle the question we computed, for A_{FE} and A_{all}, the exact worst-case threshold $\delta^*(A)$: the largest δ such that $A(p+e) = L$ for every $L \in W_g(Q)$ and every error vector with $|\, e_j \,| \leq \delta$. The loss of the operator is then

$$loss(A) = \frac{\delta_{cong}}{\delta^*(A)}. \tag{9}$$

Since $\hat{c}_k$ depends only on $\hat{p}_1, \ldots, \hat{p}_k$, the error $Q^{k/2}(\hat{c}_k - c_k)$ is a polynomial in $e_1, \ldots, e_k$. Two independent computations bracket δ^*:

- **Sufficiency (rigorous):** the exact expansion of that polynomial, bounded term by term by $\sum |\, coeff \,| \cdot \delta^{|\alpha|} < \frac{1}{2}$ for every member and every rounded coefficient.
- **Failure (constructive):** a vertex enumeration and bounded optimisation over the error box, yielding an explicit record on which the operator returns a wrong polynomial.

The two bounds agree to twelve significant digits in every case, so the values below are exact. Every failure witness was confirmed by running the operator: it fails at $1.001 \cdot \delta^*$ and succeeds at $0.999 \cdot \delta^*$.

5.2 Closed form for genus 2

For $g = 2$ the coefficients rounded by A_{FE} are $Q^{1/2}c_1 = -Q^{1/2}p_1$ and $Qc_2 = \frac{Q}{2}(p_1^2 - p_2)$. Maximising the second over the error box gives $\frac{Q}{2}(2 \mid p_1 \mid \delta + \delta^2 + \delta)$, attained at a vertex. With $P = max \mid p_1 \mid = \lfloor 4\sqrt{Q} \rfloor / \sqrt{Q}$, the exact threshold is

$$\delta^*(A_{FE}) = min(\frac{1}{2\sqrt{Q}}, \delta_2), \qquad Q\delta_2^2 + Q(2P+1)\delta_2 - 1 = 0. \tag{10}$$

Since $P \to 4$ and $\delta_{cong} = 1/Q$ for $Q \geq 4$, the loss tends to 9. The extremal member is always the polynomial of largest $\mid b_1 \mid$, of the form $(1 \mp \sqrt{Q}T)^{2g}$, perturbed with alternating signs.

5.3 Results

g	Q	$\mid W_g(Q) \mid$	δ_{cong}	$\delta^*(A_{FE})$	Stated loss A_{FE}	True loss A_{FE}	Stated loss A_{all}	True loss A_{all}
1	3	7	0.28868	0.28868	6.21	**1.00**	—	—
1	25	21	0.10000	0.10000	6.07	**1.00**	—	—
1	128	45	0.04419	0.04419	6.03	**1.00**	—	—
2	2	35	0.35355	0.074282	47.5	**4.76**	94.6	**17.3**
2	3	63	0.28868	0.041823	58.0	**6.90**	173.5	**38.5**
2	4	101	0.25000	0.027693	66.9	**9.03**	266.9	**68.8**
2	5	129	0.20000	0.024450	66.9	**8.18**	333.5	**76.3**
2	7	207	0.14286	0.016658	66.8	**8.58**	466.8	**113.1**
2	9	311	0.11111	0.012329	66.8	**9.01**	600.1	**154.6**
2	25	1371	0.04000	0.0044423	66.7	**9.00**	1666.7	**429.2**
3	2	215	0.35355	0.015379	314.0	**23.0**	—	—
3	3	677	0.28868	0.0064359	470.8	**44.9**	—	—
3	5	2953	0.13416	0.0028620	470.6	**46.9**	—	—

Table 1. Stated versus true worst-case loss, loss = δ_{cong}/δ^*. "Stated" is δ_{cong} divided by the sufficient bound of [CP, Thm 3.2, Prop. 6.1], as reported in [RT]; "true" is computed exactly as in §5.1.

Three observations follow.

- **Genus 1: no loss.** Rounding $b_1 = -s_1$ is already the optimal decoder, so A_{FE} attains δ_{cong} exactly; the stated factor 6 is entirely the slack of the sufficient bound.
- **Genus 2 and 3:** the true loss of A_{FE} is about 7–10 times smaller than stated (≈9 against ≈67 for $g = 2$; 23–47 against 314–471 for $g = 3$). The same holds for A_{all}.
- **Consistency with [RT]'s own data.** Under random uniform noise [RT, Table 3], A_{FE} first fails at $0.179 \cdot \delta_{cong}$ for $g = 2$ — a loss of about 5.6, in line with the exact values above and not with $g \cdot 2^{2g} \cdot H_{2g} \approx 67$.

The accurate statement is: *the sufficient bound of [CP] is loose by up to $g \cdot 2^{2g} \cdot H_{2g}$; the true worst-case loss of Newton-then-round is 1 for $g = 1$, ≈9 for $g = 2$ and ≈23–47 for $g = 3$ ($Q \leq 5$).* The qualitative point survives: Newton-then-round is suboptimal, and its loss grows with g.

6 Statement 4: record versus reconstruction

> *"…in the reconstruction rather than in the record."*

Our paper. [CP, §1.3] separates the error sources — trace estimation, Newton reconstruction and final rounding — without using the terms "record" and "reconstruction layer".

The commentator. In the vocabulary of Return Theory (layers L1 and L2), the record of the first g traces retains every distinction down to δ_{cong}, so any shortfall belongs to the reconstruction operator.

Assessment. Correct in substance, but not additional to Statements 1 and 2. That the record separates all members down to δ_{cong} while a particular operator needs finer precision is exactly Theorem 7.3 together with §5. The RT terminology is a framing choice: none of the mathematics in Sections 6–7 of [RT] depends on it, and the results can be cited without adopting it. The size of the reconstruction loss is that of Table 1, not $g \cdot 2^{2g} \cdot H_{2g}$.

7 Statement 5: Research Question 8.1

> *"This bears on your RQ 8.1."*

Our paper. Research Question 8.1 asks, for a specified family of curves or abelian varieties, for "the precision and auxiliary arithmetic information sufficient to reconstruct its integral Frobenius polynomial *efficiently*".

The commentator. [RT, Remark 7.7] states that the result settles the precision question for the trace route.

Assessment. Correct, and the result bears on both halves of RQ 8.1.

- **Precision: settled** for the class $W_g(Q)$ and for curves of genus 1 and 2. Integrality and the functional equation are sufficient auxiliary information; δ_{cong} is sufficient and necessary.
- **Efficiency: settled negatively for the trace route.** Precision of order $Q^{-g/2}$ requires $\Theta(Q^{g/2})$ coherent calls per trace under amplitude estimation, which is exponential in $g \log Q$. Classical methods (Schoof–Pila for fixed g; Kedlaya; Harvey) remain superior, as [RT, Remark 7.7] concedes.
- **Open:** exact thresholds for curve-realised classes of genus $g \geq 3$, and the constructive-oracle question in the second paragraph of [CP, §8.1].

8 Summary

#	Statement in the comment	Our paper [CP]	Assessment
1	Exact threshold δ_{cong} for every g and Q	Not derived; left open (caveat to Thm 3.4; remark after Prop. 6.1)	**Commentator correct** on $W_g(Q)$; for curves proved only for $g \leq 2$
2	Congruence decoder attains it	Not proposed	**Commentator correct**
3	Newton-then-round loses $g \cdot 2^{2g} \cdot H_{2g}$	Bounds stated as sufficient only; no optimality claim	**Commentator incorrect**: true loss 1 / ≈9 / ≈23–47 for g = 1 / 2 / 3
4	Loss lies in reconstruction, not record	Error sources separated	Correct in substance; terminology optional

#	Statement in the comment	Our paper [CP]	Assessment
5	Bears on RQ 8.1	Question posed	**Correct**: precision settled; trace route not efficient

Table 2. The five statements of the comment against the original research.

Requests and next steps

- We will cite [RT, Theorem 7.3 and Lemma 7.2] in the next revision of [CP].
- We ask the commentator to restate Corollary 7.6 and the abstract of [RT] as: "the sufficient bound of [CP, Thm 3.2, Prop. 6.1] is loose by up to $g \cdot 2^{2g} \cdot H_{2g}$", and to report the true worst-case loss of the operator.
- In our next revision we will state explicitly that the $d = 64$ benchmark of [CP, §2.1] uses the repeated-block spectrum of Proposition 6.3; [RT, Remark 5.3] reports that double-precision Newton deviates from numpy.poly by 0.15 at $d = 64$ on Haar-random spectra.

9 Verification methods and reproducibility

Replay of the commentator's package

- All 38 SHA-256 checksums in SHA256SUMS match. The code was inspected before execution: it has no network access and writes only within its own results and generated-TeX folders.
- python -I code/run_all.py completed all ten stages (Python 3.13, NumPy 2.5, SciPy 1.18). Every result file and every generated TeX table is byte-identical to the shipped versions; only the run manifest (timings) and the PDF figures (embedded timestamps) differ.

Independent checks

- **Weil classes:** exact enumeration via the trace polynomial $h(y)$, $y = T^{-1} + QT$, using $V_n = yV_{n-1} - QV_{n-2}$; membership iff h has g real roots in $[-2\sqrt{Q}, 2\sqrt{Q}]$, decided by exact factorisation and real-root isolation.
- **Exact threshold:** brute-force δ_{sep} over all pairs; 16/16 cases equal δ_{cong}.
- **Witness pairs:** exact rational power sums for $g \leq 12$; root moduli by numerical root-finding.
- **Anchor curve:** independent point count of $y^2 = x^5 - x$ over F_{3^m}, $m = 1, \ldots, 4$, with an independently constructed field: $N_m = 4,6,28,110$, $s_m = 0,4,0,-28$, $L(T) = 1 - 2T^2 + 9T^4$, in agreement with [CP, Prop. 6.2].
- **Operator thresholds:** exact polynomial expansion of coefficient errors (rigorous sufficiency) and constructive failure witnesses, agreeing to 12 digits; witnesses validated by direct execution.

The validation scripts (frobenius_threshold_verification.py and operator_failure_witness_check.py), their JSON outputs and a README are released with this paper under the MIT License.

References

[1] R. Cooperman and Petrunina, Quantum Trace Estimation and Frobenius Polynomial Reconstruction, revised 26 September 2026.

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